Shelf Voice — 06 UI/UX and Design

Oct 27, 2026 · @Aradhya Chaudhary

Status and open decisions

Version 1.0, approved by the owner on Oct 27. All five design decisions are final; changes from here go through this table. The Signboard tokens below are now the design baseline.

#	Decision	Options (recommendation %)	Why the top option	Owner's choice
U1	Visual direction	Signboard: painted shop-sign green and turmeric (60%); Khata: account-ledger red and ink blue (25%); Clean utility: neutral with one brand colour (15%)	Drawn from the kirana street itself, readable in sunlight, and distinct from generic AI-styled apps	Signboard
U2	Owner app default language	Follow the phone's language, toggle always visible (65%); always start in Hindi (25%); always start in English (10%)	Respects what the owner already chose on their phone; one tap to switch	Follow phone language
U3	Owner app navigation	Bottom tab bar with a large centre mic button (70%); single home screen with cards (30%)	Mic is always one thumb-tap away; four tabs match the four jobs	Bottom tabs + centre mic
U4	Storefront product layout	List rows grouped by category chips (65%); photo card grid (35%)	Most kirana items will not have photos at first; lists show more items per screen	List rows + category chips

#	Decision	Options (recommendation %)	Why the top option	Owner's choice
U5	Visual mockups before building	Design system + mockups of 6 key screens as editable design artifacts, right after this document (70%); design directly in code (30%)	The team builds against one agreed picture; mockups also feed the demo video	Yes: design system + 6 screens, made once this document is approved

Design objectives and principles

The owner should be able to run Shelf Voice with one thumb, at a busy counter, in daylight, without reading a paragraph.

Objectives:

1. Speak first: every main job starts from the mic; typing is always possible but never required.
2. Confirm, don't trust: AI results appear as clear, editable proposals with one obvious confirm button.
3. Numbers at a glance: stock, days left and reorder quantity are the biggest text on their screens.
4. Customers in 15 seconds: storefront and review page load fast and need zero sign-up.

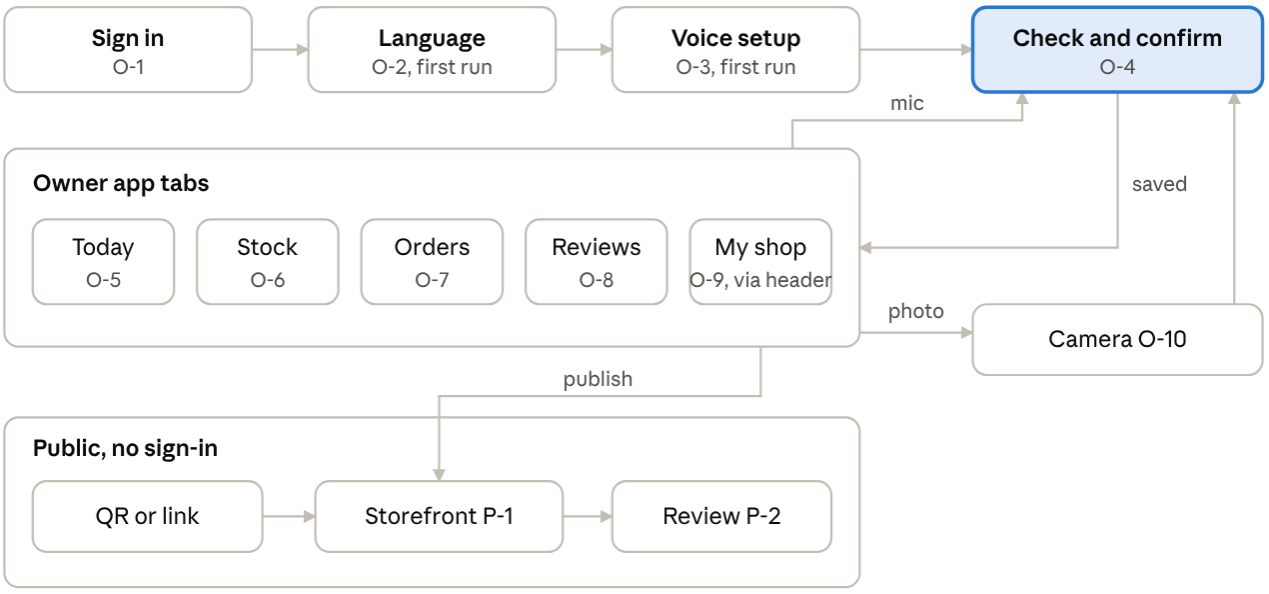
Principles:

- Icon plus word, always. Labels never rely on icon or colour alone, which helps owners who read slowly and meets accessibility rules.
- One primary action per screen, in the thumb zone at the bottom.
- Plain everyday words in Hindi and English ("Stock khatam", "Order karo"), never system terms like "proposal" or "SKU" in the interface.
- Large targets: at least 48 by 48 px, with 8 px between them.
- Show what the AI heard: the transcript sits above every proposal, so the owner can see why it suggested something.
- Calm by default: one memorable element (the mic), everything else quiet.

Information architecture

After a one-time setup, the owner lives in four tabs (Today, Stock, Orders, Reviews) with the mic in the centre slot; My shop opens from the shop name on Today. Every voice or photo action returns through the same Check and confirm screen.

Owners land on five tabs; customers only see two public pages



screen map · 10 owner screens, 2 public

Customers never see the owner app: they reach the storefront from a shared link or QR, and the review page from the counter QR or a link at the bottom of the storefront. Reviews they leave appear in the owner's Reviews tab.

Screen inventory

Thirteen screens in total: ten for the owner, two public, one for the team.

ID	Screen	Purpose	Key elements	PRD
O-1	Sign in	Get in fast	"Continue with Google", "Try the demo shop", language switch	FR-01
O-2	Language	First-run choice	Hindi, English; preselected from phone language (U2)	NFR
O-3	Tell us about your shop	First-run voice setup	Big mic, three example prompts ("shop name, timings, what you sell"), 2-minute timer	FR-02

ID	Screen	Purpose	Key elements	PRD
O-4	Check and confirm	Review any AI proposal	What we heard (transcript), list of changes with edit and remove, Confirm all	FR-03, 05, 06, 11
O-5	Today (home)	Daily overview	Today's sales total, items running low, new reviews, centre mic	FR-13
O-6	Stock	Catalog with stock levels	Search, category chips, each row: name, price, stock, in/out toggle	FR-05
O-7	Order list	What to restock	Rows sorted by days left: quantity to order, one-line reason, review flag	FR-07, 09
O-8	Reviews	Customer voice feed	Sentiment icon, transcript, items mentioned, play audio (30 days), hide	FR-09
O-9	My shop	Storefront and settings	Live preview, Publish switch, QR download, share link, hours, delete shop	FR-04, 16
O-10	Camera	Bill or shelf photo	Viewfinder guide, retake, use photo	FR-10, 11
P-1	Storefront	Customer browsing	Shop name, open-now status, hours, categories, item list, call and WhatsApp order buttons	FR-04
P-2	Leave a voice review	Customer feedback	Consent notice, record button with 15-second ring, thank-you	FR-08
A-1	Admin	Team monitoring	Credit used, AI success and latency, product counts (document 05)	NFR

Voice interaction design

The mic is the one memorable element: a large round button that always behaves the same way, from every screen.

Mic button behaviour:

State	Looks like	Says	Haptic
Idle	Filled primary circle, mic icon, 72 px	"Bolo" / "Speak" under it	None
Recording	Ring fills clockwise as the time limit approaches; live level bars inside	Timer "0:12 / 0:30"; "Tap to stop"	Short buzz on start
Uploading and thinking	Ring spins slowly; button disabled	"Sun raha hoon..." / "Listening..." then "Samajh raha hoon..." / "Understanding..."	None
Done	Moves to Check and confirm (O-4)		Short buzz
Failed	Button returns to idle; message above it	What failed and what to do ("Couldn't hear you. Try again closer to the phone.")	Double buzz

Interaction rules:

- Tap to start and tap to stop (not press-and-hold), because owners often record while handling goods.
- Recording stops by itself at the limit: 2 minutes for setup, 30 seconds for commands, 15 seconds for customer reviews.
- The first time, a 3-line tip shows what to say, with examples in the owner's language.
- Microphone permission is requested only when the mic is first tapped, with a one-line reason before the browser prompt.

Check and confirm pattern (O-4):

1. Top: "What we heard" with the transcript, collapsed to 2 lines with "show all".
2. Middle: one card per change, written as a plain sentence ("Maggi: price Rs 14 to Rs 15"), with edit and remove.
3. Anything unclear is marked "Please check" in amber and never pre-selected.
4. Bottom, in the thumb zone: "Confirm 5 changes" (the count updates as cards are removed) and "Cancel".
5. After confirm: toast "5 changes saved" with Undo for 10 seconds.

Visual direction and tokens

The recommended direction, Signboard, borrows from hand-painted kirana shop signs: deep leaf green, turmeric yellow and heavy rounded lettering, on clean white for daylight readability.

The three options (U1):

Direction	Idea	Palette	Type	Risk
Signboard (60%)	Painted shop signboards and shutters	Leaf green #0B6E4F, turmeric #F2B705, ink #1B1F1A, white #FFFFFF, leaf-grey #F3F5F1	Baloo 2 for shop names and big numbers; Mukta for interface text (both cover Devanagari and Latin)	Turmeric must never carry text on white (low contrast); used only as a fill behind dark text
Khata (25%)	The red-bound account ledger every shop keeps	Ledger red #A4161A, ink blue #1D3557, rule grey #E3E7EF, white	Tiro Devanagari headings, Hind body	Ruled-ledger look can feel dense on small phones
Clean utility (15%)	Neutral app with one brand colour	Blue #2457D6, greys, white	Inter + Noto Sans Devanagari	Looks like every other app; weaker identity

Tokens for Signboard (chosen Oct 27):

Token	Value	Use
color.primary	#0B6E4F	Mic, primary buttons, links (6.2:1 on white)
color.accent	#F2B705	Highlights, "low stock" fill behind ink text
color.ink	#1B1F1A	Body text
color.ink-muted	#5B6459	Secondary text (6.2:1 on white)
color.surface	#FFFFFF	Page background
color.surface-muted	#F3F5F1	Grouped sections, chips

Token	Value	Use
color.border	#D5DBD3	Dividers, inputs
color.danger	#B42318	Out of stock, delete, errors (6.6:1)
color.check	#9A5B00 on #FFF4D6	"Please check" items
type.display	Baloo 2, 600-800	Shop name, big numbers
type.ui	Mukta, 400-700	All interface text
type.scale (px)	14, 16, 20, 24, 32, 40	Body is 16; key numbers 32-40
space (px)	4, 8, 12, 16, 24, 32, 48	Only these values
radius (px)	6 for cards and inputs; full round for mic and chips	Hierarchy: round means "tap to speak or filter"
shadow	None, except the mic button	Flat, sunlight-readable

Dark mode: supported through the same tokens with a dark surface, but light is the default because counters are bright.

Component library

Base components come from shadcn/ui (Radix primitives, accessible by default) restyled with our tokens; seven product components are built on top and are the only new ones.

From shadcn/ui: Button, Input, Textarea, Select, Switch, Tabs, Sheet (bottom drawer), Dialog, Toast (Sonner), Badge, Skeleton, Separator. Icons: Lucide, always paired with a text label.

Product components:

Component	Used on	Behaviour
MicButton	O-3, O-5, all tabs, P-2	States from "Voice interaction design"; time limit passed as a prop; announces state changes to screen readers
TranscriptPanel	O-4	Collapsible "What we heard" text
ChangeCard	O-4	One proposed change as a sentence; edit, remove; amber "Please check" variant
StockRow	O-6, P-1 (simplified)	Name, unit, price, stock count or in/out badge; tap to edit

Component	Used on	Behaviour
ReorderRow	O-7	Days left (large), order quantity, reason line, review-flag icon
ReviewCard	O-8	Sentiment icon and word, transcript, item chips, play button (hidden after 30 days), hide
OpenNowBadge	P-1, O-9	"Open now" / "Closed, opens 7 AM" computed from hours in the shop's time zone

Each product component has Vitest tests for its states and an axe check in its E2E screen.

Interaction states

Every screen designs its empty, loading, error and offline state up front; an empty screen always tells the owner what to do next.

Screen	Loading	Empty	Error	Success
Today (O-5)	Skeleton cards	"No sales logged today. Tap the mic and say what you sold."	"Couldn't load today's numbers" + Retry	n/a
Stock (O-6)	Skeleton rows	"Your shop has no products yet. Tell us what you sell." + mic	Same pattern	Toast "Saved" after edits
Order list (O-7)	Skeleton rows	"Nothing to order right now. Stock looks fine."	Same pattern	n/a
Order list, first 14 days	n/a	"Smart order list starts after 2 weeks of sales. Until then we warn you when stock is low."	n/a	n/a

Screen	Loading	Empty	Error	Success
Reviews (O-8)	Skeleton cards	"No reviews yet. Print your QR and keep it at the counter." + Download QR	Same pattern	n/a
Check and confirm (O-4)	"Listening… Understanding…"	"We didn't hear any changes. Try again?"	What failed + "Your recording is saved" + Retry	"5 changes saved" + Undo
Storefront (P-1)	Server-rendered, no spinner	"This shop is adding products. Call them for now."	404 page with shop-finder link if unpublished	n/a
Review (P-2)	Ring while uploading	n/a	"Couldn't send. Tap to try again."	"Thank you! The shop will hear this."

Offline: a slim bar at the top reads "Offline. Recordings will send when you're back online." The mic keeps working and queued clips show a small clock icon until sent.

AI paused (budget or outage): a banner reads "Voice is resting for now. You can still edit by tapping." and the mic shows a muted state.

Storefront and review page design

The storefront leads with the shop itself, the way a painted signboard does: its name in large Baloo 2, whether it is open right now, and two buttons to reach it.

Storefront layout at 360 px (U4 list option):

```
+----------------------------------+
|  SHARMA KIRANA STORE             |  shop name, display type
|  Open now - closes 10 PM         |  OpenNowBadge
|  Shastri Nagar - 1.2 km          |  address (no map in MVP)
|  [ Call ]   [ Order on WhatsApp ]|  primary actions
+----------------------------------+
| (All) (Atta & Rice) (Snacks) ... |  category chips, sticky
+----------------------------------+
| Maggi 2-Minute Noodles 70g       |
| Rs 14             In stock       |  StockRow
| Aashirvaad Atta 5kg              |
| Rs 245            Out of stock   |  greyed, still listed
| ...                              |
+----------------------------------+
| Leave a voice review  (mic)      |  link to P-2
+----------------------------------+
```

- "Order on WhatsApp" opens a prefilled message: shop name, a list of items the customer ticked (an optional tick per row), and "Is this available?". No cart, no payment (PRD D4).
- Out-of-stock items stay visible but greyed, so customers know the shop normally carries them.
- Prices only appear if the owner set them; stock counts are never shown publicly, only in or out.
- Open Graph image: shop name on leaf green with the turmeric bar, so WhatsApp link previews look like a signboard.

Review page (P-2) sequence:

1. Shop name, one line: "Tell [shop] how they did, in 15 seconds."
2. Consent notice (document 03) with "I agree, record" as the only button.
3. MicButton with the 15-second ring; "Tap to stop".
4. "Thank you! The shop will hear this." plus a link back to the storefront.

No name, phone or rating stars are asked for; the voice is the whole review.

Responsive behaviour, accessibility and motion

The owner app is designed for a 360 px phone first; desktop is a centred column, not a different layout.

Width	Owner app	Storefront
320-479 px	Bottom tab bar, centre mic, single column	Single column, sticky chips
480-1023 px	Same, max content width 560 px	Same, max width 640 px
1024 px and up	Left rail replaces bottom bar; mic stays bottom-right	Two columns: shop info left, products right

Accessibility (WCAG 2.1 AA):

- Text contrast at least 4.5:1, large numbers at least 3:1; turmeric never carries text on white.
- Every control reachable by keyboard with a visible 2 px focus ring in leaf green.
- MicButton has an accessible name that changes with state ("Start recording", "Stop recording, 12 seconds").
- Status changes (recording started, changes saved, errors) announced through a polite live region.
- `lang="hi"` or `lang="en"` set on the page and on mixed-language fragments, so screen readers pronounce Hindi correctly.
- Text scales to 200% without horizontal scrolling.

Motion:

- One signature motion: the mic's ring filling while recording. It answers the owner's action and shows time left.
- Elsewhere only feedback motion: a 150 ms fade for toasts and sheet slides.
- With reduced motion turned on, the ring shows a number countdown instead and slides become instant.

Design acceptance criteria

A screen is design-complete only when every box below is true on a real Android phone and in the automated checks.

- [ ] Uses only the token values in this document (no stray colours, sizes or spacing)
- [ ] Loading, empty, error and success states built as specified
- [ ] Every control is at least 48 px tall with a text label
- [ ] Primary action sits in the bottom thumb zone on phones
- [ ] No horizontal scroll at 320, 360, 768 and 1280 px
- [ ] axe-core reports zero serious or critical issues

- [ ] Keyboard-only walkthrough works with a visible focus ring
- [ ] All interface text exists in Hindi and English, and no string is cut off in either language
- [ ] Reduced-motion setting respected
- [ ] Storefront passes the Lighthouse targets in document 04
- [ ] A teammate who did not build the screen completes its task in one try without help
- [ ] At least one real shop owner has used the setup and confirm screens and their feedback is recorded